

SECR2043 OPERATING SYSTEMS

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 Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write **"No output"** instead.

1.	Display your current working directory.	
	Command	Output
	<code>damiya@secr2043:~\$ pwd</code>	<code>/home/damiya</code>
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	<code>damiya@secr2043:~\$ ls -a ~</code>	<pre> . .bash_logout .profile file2.txt 'nano hello.txt' practice.txt workspace .. .bashrc combined.txt filelist.txt 'nano sample.txt' sample.txt .bash_history local example.txt linux-lab newshare share </pre>
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	<code>damiya@secr2043:~\$ mkdir ~/os_lab</code>	No output
4.	Change your current working directory to "os_lab".	
	Command	Output
	<code>damiya@secr2043:~\$ cd ~/os_lab</code>	<code>damiya@secr2043:~/os_lab\$</code>

5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ echo "Hello, world!" > hello.txt</pre>	No output
6.	Display the content of "hello.txt".	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ cat hello.txt</pre>	<pre>Hello, world!</pre>
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ cp hello.txt hello_copy.txt</pre>	<pre>damiya@secr2043:~/os_lab\$ cat hello_copy.txt Hello, world!</pre>
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ mv hello_copy.txt hello_bak.txt</pre>	No output
9.	Delete "hello.txt".	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ rm hello.txt</pre>	No output
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ ls -R -lh ~</pre>	<pre> /home/damiya: total 48K -rw-r--r-- 1 damiya damiya 156 Apr 22 13:03 combined.txt -rw-r--r-- 1 damiya damiya 8 Apr 25 12:30 example.txt -r--r--r-- 1 damiya damiya 78 Apr 22 13:00 file2.txt -rw-r--r-- 1 damiya damiya 340 Apr 22 13:01 filelist.txt drwxr-xr-x 2 damiya damiya 4.0K Apr 22 12:45 linux-lab -rw-r--r-- 1 damiya damiya 45 Apr 22 13:12 'nano hello.txt' -rw-r--r-- 1 damiya damiya 726 Apr 22 13:14 'nano sample.txt' drwxr-xr-x 2 damiya damiya 4.0K Apr 25 12:33 newshare drwxr-xr-x 2 damiya damiya 4.0K Apr 25 12:48 os_lab -rw-r--r-- 1 damiya damiya 725 Apr 22 13:21 practice.txt -rw-r--r-- 1 damiya damiya 834 Apr 22 13:20 sample.txt drwxr-xr-x 2 damiya damiya 4.0K Apr 25 12:30 share drwxr-xr-x 2 damiya damiya 4.0K Apr 22 13:27 workspace /home/damiya/linux-lab: total 0 /home/damiya/newshare: total 0 -rw-r--r-- 1 damiya damiya 0 Apr 25 12:33 example.txt -rw-r--r-- 1 damiya damiya 0 Apr 25 12:33 textfile01.txt </pre>

		<pre> /home/damiya/os_lab: total 4.0K -rw-r--r-- 1 damiya damiya 14 Apr 25 12:46 hello_bak.txt /home/damiya/share: total 0 -rw-r--r-- 1 damiya damiya 0 Apr 25 12:21 textfile01.txt -rw-r--r-- 1 damiya damiya 0 Apr 25 12:30 textfile04.txt /home/damiya/workspace: total 4.0K -rw-r--r-- 1 damiya damiya 77 Apr 22 13:27 hello_world.c </pre>
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

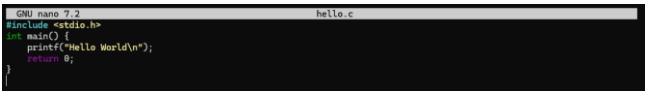
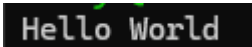
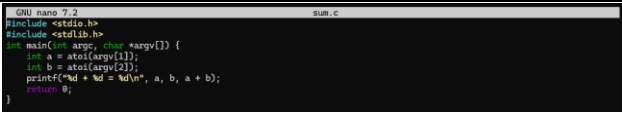
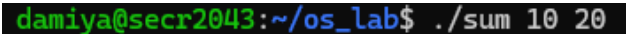
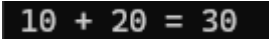
1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ touch hello.txt</pre>	No output
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ echo "Hello World" > hello.txt</pre>	No output
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ echo "This is a test" >> hello.txt</pre>	No output
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ cat hello.txt</pre>	<pre> Hello World This is a test </pre>

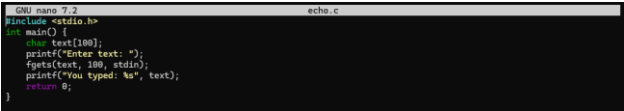
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ echo "Goodbye World" > bye.txt</pre>	No output
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ cat hello.txt bye.txt</pre>	<pre>Hello World This is a test Goodbye World</pre>
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ cat hello.txt bye.txt > greeting.txt</pre>	No output
8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ cat greeting.txt</pre>	<pre>Hello World This is a test Goodbye World</pre>
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
		No output

10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	<pre>damiya@secr2043:~/os_lab\$ cat greeting.txt</pre>	<pre>Hello Damiya This is a test Goodbye Damiya</pre>
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
		No output
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
		No Output
3.	Run the executable file "hello" and display its output.	
	Command	Output
		
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
		No output
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
		No output
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
		
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	

	Command	Output
	<code>damiya@secr2043:~/os_lab\$./sum -5 15</code>	<code>-5 + 15 = 10</code>
8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
		No output
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	<code>damiya@secr2043:~/os_lab\$ gcc echo.c -o echo</code>	No output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	<code>damiya@secr2043:~/os_lab\$./echo</code>	<code>Enter text: Hello World You typed: Hello World</code>
	Total Mark:	